$$\Phi = \frac{a_2}{2}x^2 + b_2xy + \frac{c_2}{2}y^2$$

$$\sigma_x = \frac{\partial^2\Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2\Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2\Phi}{\partial x\,\partial y} - Xy - Yx$$

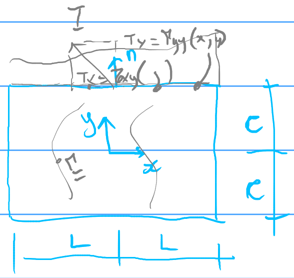
$$\nabla^4\Phi = \frac{\partial^4\Phi}{\partial x^4} + 2\frac{\partial^4\Phi}{\partial x^2\partial y^2} + \frac{\partial^4\Phi}{\partial y^4} = 0$$

Material elástico lineal
(Ley de Hooke)

$\varepsilon =$

$$\varepsilon_x = \frac{\sigma_x - \nu\sigma_y}{E}, \quad \varepsilon_y = \frac{\sigma_y - \nu\sigma_x}{E}, \quad \gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\nu)\tau_{xy}}{E}$$

$$\frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2} = 2\frac{\partial^2 e_{xy}}{\partial x\,\partial y},$$



$T = \underline{\underline{\Upsilon}} \cdot n$ $\quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$